CHAPTER 5

Trigonometric functions

S75, S76, S78, S79

Before you start · Trigonometric Circle

Without a calculator, in six minutes. Respond briefly. Mobilize radians, signs and diversion.
choose the targeted review indicated under the questions.

1. Convert 180 ° In radians.
2. Give sin 0 and cos 0.
3. Give the sign of sin x on $]0;\pi[$.
4. Solve cos $x = 0$ on $[0;2\pi]$.
5. Give cos $(\pi/3)$.
6. Give the sin period.

Responses

π ; 0 and 1 ; positive ; $\frac{\pi}{2}$ and $\frac{3\pi}{2}$; $1/2$; 2π.

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Take a turn again on the circle; R85. **2.** Read abscissa and ordered from point (1.0). **3.** Read the sign of order on the upper half circle; R86. **4.** Identify the two points of zero abscisse. **5.** Use a equilateral triangle cut in half. **6.** Make a complete tour and compare two ordered; R85. R questions serve as support for the teacher, even if their collective ritual is planned later.

Objectives

Use parity and periodicity; drift sine and cosine; solve equations and inequities; study a periodic function.

Course notes

Circle and functions

On the trigonometric circle, the point identified by an angle x in radians a for coordinates $(\cos x, \sin x)$. For all real x,

$$\cos^2 x + \sin^2 x = 1, \quad -1 \leq \cos x, \sin x \leq 1.$$

Cosine is paired: $\cos(-x) = \cos x$. Sine is odd: $\sin(-x) = -\sin x$. The two functions are: 2π-periodic.

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1

Derivatives and variations

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x.$$

$$(\sin(ax+b))' = a\cos(ax+b), \quad (\cos(ax+b))' = -a\sin(ax+b).$$

On $[-\pi, \pi]$, cosine grows on $[-\pi, 0]$ of -1 to 1, then decreases on $[0, \pi]$ from 1 to -1. Sine decreases on $[-\pi, -\frac{\pi}{2}]$, grows on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, then decreases on $[\frac{\pi}{2}, \pi]$.

Justification. Read the sign of sine and cosine on the circle and then apply the derivative link·variations. The periodicity prolongs the curves; it does not exempt from studying a length interval sufficient.

Equations and inequities

If $|a| > 1$, $\cos x = a$ and $\sin x = a$ no real solution. For a known angle θ,

$$\cos x = \cos\theta \iff x = \theta + 2k\pi \text{ or } x = -\theta + 2k\pi,$$

$$\sin x = \sin\theta \iff x = \theta + 2k\pi \text{ or } x = \pi - \theta + 2k\pi, \quad k \in \mathbb{Z}.$$

The solutions are then restricted to the required range and duplicates are eliminated.

Examples on $[-\pi, \pi]$. $\cos x = \frac{1}{2}$ gives $x = \pm\frac{\pi}{3}$; $\sin x = \frac{1}{2}$ gives $x = \frac{\pi}{6}$ or $\frac{5\pi}{6}$. $\cos x \leq \frac{1}{2}$ gives $[-\pi, -\frac{\pi}{3}] \cup [\frac{\pi}{3}, \pi]$; $\sin x \geq \frac{1}{2}$ gives $[\frac{\pi}{6}, \frac{5\pi}{6}]$.

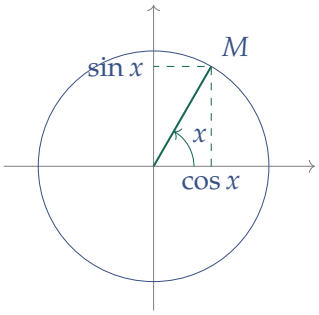
A trigonometric compound

To solve $\cos(2x) = \frac{1}{2}$ on $[-\pi, \pi]$, pose $t = 2x \in [-2\pi, 2\pi]$, solve for t, then divide by 2. $x \in \{-\frac{5\pi}{6}, -\frac{\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}\}$.

Optimization. For $f(x) = \sin x + \frac{1}{2}\sin(2x)$ on $[0, \pi]$, $f' = \cos x + \cos(2x) = 2\cos^2 x + \cos x - 1 = (2\cos x - 1)(\cos x + 1)$. The function grows up to $\frac{\pi}{3}$ then decreases; maximum $\frac{3\sqrt{3}}{4}$.

Duplication identities used in exercises. For all real x, $\sin(2x) = 2\sin x\cos x$ and $\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1$. These identities are recalled here so that their employment in problems are not implicit.

Curve symmetries. The cosine parity corresponds to a symmetry with respect to the axis of The periodicity extends to a central symmetry of the original centre. each vector translation curve $(2\pi, 0)$.



The coordinates of the point of the circle are $(\cos x, \sin x)$.

From circle to study of function

One angle oriented, two coordinates

First use: S75.

Definition · Sine and cosine

On the unit circle oriented in the direct direction, the point of angle x has for coordinates $(\cos x; \sin x)$. Add 2 $k\pi$ with $k \in \mathbb{Z}$ gives the same point. A negative measure corresponds in the opposite sense; it is not a negative length.

Example Solved · Read an equation and an inequation

On $[-\pi; \pi]$, $\cos x = 1/2$ corresponds to the two points of abscisse 1/2: $x = \pm\pi/3$. Equation $\cos x \leq 1/2$ selects the arc to the left of the vertical right of the abscisse 1/2: $[-\pi; -\pi/3] \cup [\pi/3; \pi]$. The boundaries belong to the solution because inequality is wide.

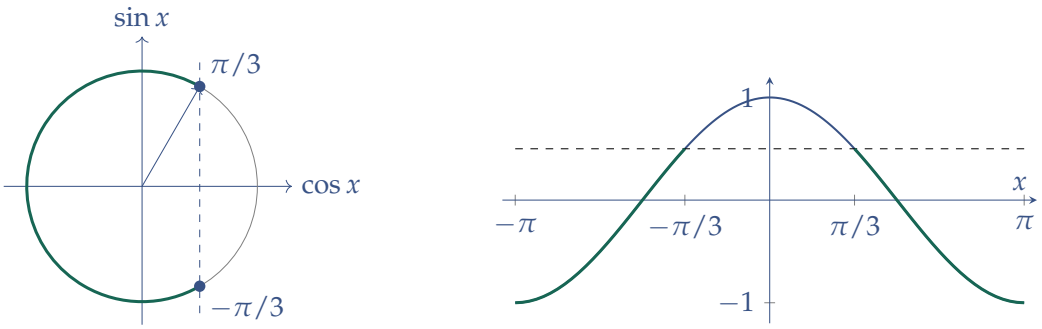


Figure 5.1. Same condition $\cos x \leq 1/2$: arc to the left of the vertical on the circle, curve below level 1 /2 à Right.

A1 · Immediate application Solve $\sin x = 1/2$ on $[0; 2\pi[$then $\sin x \geq 1/2$ on the same Indicate solutions on a circle and curve.

5.2 Read amplitude and period

First use: S76.

Example resolved · A staggered oscillation

For $h(t) = 3 + 2\sin(4t)$, the values are between 1 and 5 : the average level is 3, The smallest positive period is 2 $\pi/4 = \pi/2$. Number 2 π remains one period, but is not the smallest.

A2 · Partial guidance For $g(t) = 1 - 3\cos(2t)$, give minimum, maximum, amplitude and more In order to determine the values of t in $[0;\pi]$ where the extrema are affected.

A3 · Refuter "The smallest positive period of $\sin^2 x$ is 2 π, such as that of $\sin x$. » Show that a translation of π leaves the function unchanged; explain why $\pi/2$ is not a period.

5.3 The sign comes from the circle

First use: S78.

Property · Derivatives and antiderivatives

The derivatives are: $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$. For a compound, multiply by the A primal of $\cos(ax+b)$ for $a \neq 0$ is $\sin(ax+b)/a$; a antiderivative Of sin $(ax+b)$ is $-\cos(ax+b)/a$.

Example Solved · Variation over an interval

For $f(x) = x + \cos x$ on $[0;\pi]$, $f' = 1 - \sin x \geq 0$, with equality only in $\pi/2$. The function is strictly increasing: it grows strictly on either side of this isolated point and remains continues. $f'' = -\cos x$ change of sign in $\pi/2$, so the curve has inflection in $(\pi/2;\pi/2)$. A horizontal tangent is not necessarily an extreme.

A4 · Choice of strategy Study $f(x) = x - \sin x$ on $[0;\pi]$: derivative, variations, convexity. $\sin x \leq x$ on this interval. $f'(0) =$ Is it causing a change in variation?

5.4 An identity serving an area

First use: S79.

Example Solved · Area of a rectangle

One point of the first quadrant of the unit circle has for coordinates $(\cos\theta;\sin\theta)$, $0 \leq \theta \leq \pi/2$. The rectangle symmetrical with respect to the axes has as its area $A(\theta) = 4\sin\theta\cos\theta$. Like $(\sin\theta - \cos\theta)^2 \geq 0$ and $\sin^2\theta + \cos^2\theta = 1$, we have $2\sin\theta\cos\theta \leq 1$, therefore $A \leq 2$. Equality requires $\theta = \pi/4$: the optimal rectangle is a square.

A5 · Summary Determine by bypass the maximum of 4 $\sin\theta\cos\theta$ on $[0;\pi/2]$. Find the previous conclusion, then calculate $\int_0^{\pi/2} \sin\theta\cos\theta\,\mathrm{d}\theta$.

A6 · Autonomous problem Demonstrate that $\cos x = x$ has a single solution on $[0;\pi/2]$. Framework between 0.7 and 0.8 using the radian-mode calculator. precision.

A first step: Study the difference between the two members and separate existence, uniqueness and approximation.

Wait a minute at the courthouse. Explain how the same set of solutions appears as arcs on the circle and intervals on the axis of the apses of the curve.

Review postponed · S84 and then S90 Mixing boundaries, derivation, equations, areas and trigonometry in same study; repeat a resolution over another interval by controlling the bounds.

Research activity

Height of one point on a wheel · 35 min.

A wheel rotates at constant speed in a vertical plane. Its radius is 2 m and its centre is found 3 m from the ground. A point of the wheel is at its lowest position at the moment $t = 0$. Wheel makes a turn in 8 s. Its height is modelled by $h(t) = 3 - 2\cos\left(\frac{\pi t}{4}\right)$ where: t is expressed in seconds and $h(t)$ In meters.

1. Verify that the model gives the initial height announced, a period of 8 s and heights minimum and maximum expected.
2. Calculate $h'(t)$, then study the variations in h on $[0;8]$.
3. Determine every moment $t \in [0;8]$ for which the height is 4 m.
4. Determine the total time during this round during which the point is at least 4 m of the ground.

Progressive exercises

E1 · Direct application. Solve in $[-\pi;\pi]$ the sin equation $x = -\frac{1}{2}$.

E2 · Application and justification. Solve in $[-\pi;\pi]$ Iin equation cos $x \geq \frac{\sqrt{2}}{2}$.

E3 · Method to be organised. We define on $\mathbb{R}$ functions $f(x) = \cos(3x-1)$ and $g(x) = \sin^2 x$. Calculate their derivatives.

E4 · Choice of strategy. It is considered $f(x) = 2 + \sin x$ on $[0;2\pi]$. Calculate its derivative and establish its Table of changes.

E5 · Synthesis or demonstration. Calculate $\int_0^{\pi} \sin x \, \mathrm{d}x$, then the average value of the sine function on $[0;\pi]$.

Work in the manual or annal

Barbazo, p. 166, Exercise 19. In S75: question 1, to read a symmetry and a period over a period of At home: question 2, to demonstrate conjecture by identities valid for all real.

Duty at home

DM5 · Share an area · 55 min, after S79.

The function is considered f defined on $[0;\frac{\pi}{2}]$ by $f(x) = \sin x \cos x$. Its curve is represented in a orthonormal coordinate system.

1. Calculate $f'(x)$ and determine the maximum of f over this interval.

2. Solve on $[0; \frac{\pi}{2}]$ Inequation $f(x) \geq \frac{1}{4}$.

3. Calculate $\int_0^{\frac{\pi}{2}} f(x)\,\mathrm{d}x$, then the average value of f on $[0; \frac{\pi}{2}]$.

4. For $a \in [0; \frac{\pi}{2}]$, we're laying $A(a) = \int_0^a f(x)\,\mathrm{d}x$. Determine a so that the domain under the curve between 0 and a has an area equal to half of the total area under the curve between 0 and $\frac{\pi}{2}$.

QCM A · circle and functions

After S76; k means a relative integer.

For each question, only one proposal is accurate.
consult the correction.

1. $\cos(-x)$ is valid for:

A. $-\cos x$ B. $\cos x$

C. $\sin x$ D. 0

2. The smallest strictly positive period of $x \mapsto \sin(3x)$ is:

A. 2π B. 3π

C. $\frac{2\pi}{3}$ D. $\frac{\pi}{3}$

3. The solutions of sin $x = 0$ are:

A. $x = \frac{\pi}{2} + 2k\pi$ B. $x = k\pi$

C. $x = 2k\pi$ only D. $x = \pi/4$

4. The cos derivative x is:

A. $\sin x$ B. $-\sin x$

C. $-\cos x$ D. $\tan x$

5. On $[0; \pi]$, the sine is:

A. Always negative B. Positive or nil

C. Always increasing D. Still sucks

QCM B · equations and study

After S79.

For each question, only one proposal is accurate.
consult the correction.

6. Solve cos $x = 1$ on $[0; 2\pi]$ gives:

A. 0 and 2π B. π

C. $\frac{\pi}{2}$ D. One solution

7. To solve sin $x \geq \frac{1}{2}$These include:

A. choose the right arc and then intersect with the interval requested B. divide by sin x

C. use a binomial D. Forget the boundaries

8. A function of the form $a + b\cos(\omega t)$, $\omega \neq 0$ a for amplitude:

A. a B. b

C. $|b|$ D. ω

9. If $f'(x) = \cos x$, f is decreasing on:

A. $]0; \frac{\pi}{2}[$ B. $]\frac{\pi}{2}; \frac{3\pi}{2}[$

C. $]\frac{3\pi}{2}; 2\pi[$ D. $[-\frac{\pi}{2}; \frac{\pi}{2}]$

10. Two solutions that differ from 2 π represent:

A. Separate angles but same point on the circle B. A contradiction

C. Opposite sinuses D. Prohibited values

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. $\sin^2 x + \cos^2 x = 1$ for all real x.
2. $\sin x = \frac{1}{2}$ has only one solution on $[0; 2\pi]$.
3. $\sin(a + b) = \sin a + \sin b$ for all real a, b.

Review review

I can read the circle, exploit parity and period, solve with an interval and study a function. **À**
monitor: forget a ship's solution; confuse a graphic reading with proof of identity.

Corrected MCQs

1. **B.** Cosine is even: cos $(-x) = \cos x$. It's the sine that's odd.
2. **C.** The smallest positive shift in the argument equals 2 π is obtained for 3 $T = 2\pi$either $T = 2\pi/3$. 2π is also a period but is not the smallest.
3. **B.** The sine zeros are the integer multiples of π, $k \in \mathbb{Z}$. Keep only 2 $k\pi$ loses Multiple odds.
4. **B.** The cosine derivative is the opposite of the sine, the angles being radians.
5. **B.** The sine is positive or zero on the upper semicircle. It grows up to $\pi/2$ then decreases, it is Not increasing over the entire interval.
6. **A.** Both bounds 0 and 2 π are two numbers distinct from the interval, even if they correspond at the same point on the circle.
7. **A.** Identify the arc where the order is at least 1 $/2$, then apply the periodicity and the bounds of the Dividing by a sine of variable sign would be unjustified.
8. **C.** For $\omega \neq 0$, extremes are $a - |b|$ and $a + |b|$. The amplitude is half their difference, let $|b|$.
9. **B.** The cosine is negative on $]\pi/2; 3\pi/2[$; the sign of the derivative gives the decrease of f.
10. **A.** Add a complete round retains the point of the circle, as well as sine and cosine. two separate numbers remain.

True/false. True: the point of the unit circle has for coordinates $(\cos x ; \sin x)$ and sum of squares value 1. False: there is $\pi/6$ and $5\pi/6$. False: for $a = b = \pi/2$, the left limb is 0 and the right 2.

Corrected applications and support

Application 1. Equation: $\pi/6$ and $5\pi/6$. Inequation: $[\pi/6;5\pi/6]$, arc higher than the ordered level $1/2$.

Application 2. Minimum -2 in t=0 and π ; maximum 4 in $\pi/2$; amplitude 3 ; smaller period Positive π. The sign less reverses the positions of the maxima and minimum cosine.

Implementation 3. $\sin(x+\pi) = -\sin x$, so squares are equal. But values in 0 and $\pi/2$ are 0 and 1: $\pi/2$ is not a period. To prove the minimality of π, a period T>0 must check $\sin^2 T = 0$, so T is a positive integer of π.

Implementation 4. $f' = 1-\cos x \geq 0$, strictly positive on $]0;\pi]$, so f is increasing, strictly after 0. $f'' = \sin x \geq 0$, so convexity. Like f(0)=0, $x - \sin x \geq 0$. Zero derivative in one limit does not impose any change in variation.

Application 5. $A' = 4(\cos^2\theta - \sin^2\theta)$, positive before $\pi/4$ and negative after, where maximum 2. Early sin $\theta\cos\theta$ is $\frac{1}{2}\sin^2\theta$, so the integral is worth 1/2.

Implementation 6. $G(x) = \cos x - x$ is continuous, $G(0) = 1 > 0$, $G(\pi/2) = -\pi/2 < 0$; the VIT Gives a solution. $G' = -\sin x - 1 < 0$ on the interval ensures uniqueness. $G(0{,}7) \simeq 0{,}0648 > 0$, $G(0{,}8) \simeq -0{,}1033 < 0$. A dichotomy on this interval improves the framework.

Driving

Explicitly adjust the radian mode. To optimize, compare algebraic evidence and derivation: the identity explains the bound, the derivative describes the variations. a semi-open interval to have duplicates treated.

Conduct and correction

Benchmarks for the teacher

This chapter is really finished in early May, not late April.
2nd white of April. Retake the circle before curves. Use radians in the
- Controlling the bounds $-\pi$ and π : they represent the same point of the circle but are two distinct numbers if the interval contains them. The variations of compounds are deduced the complete sign of the derivative.

Corrected activity

1. $h(0) = 3 - 2 = 1$ m. Add 8 to t add 2 π to the cosine argument: 8 s is a period, and this is the smallest positive period. $-1 \leq \cos \leq 1$, $1 \leq h \leq 5$; both bounds are reached.

2. By derivation of a compound, $h'(t) = (\frac{\pi}{2})\sin(\frac{\pi t}{4})$. The argument goes through $[0, 2\pi]$. The sine is positive on $]0, \pi[$, then negative on $]\pi, 2\pi[$. Thus h grows from 1 to 5 m on $[0, 4]$, then decreases from 5 to 1 m on $[4, 8]$.

3. $h(t) = 4 \iff \cos(\frac{\pi t}{4}) = -\frac{1}{2}$. On $[0, 2\pi]$, the corresponding angles are $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$; therefore $t = \frac{8}{3}$ s or $t = \frac{16}{3}$ s.

4. The condition $h(t) \geq 4$ is equivalent to $\cos(\frac{\pi t}{4}) \leq -\frac{1}{2}$, therefore $t \in [\frac{8}{3}, \frac{16}{3}]$. The duration is $\frac{16}{3} - \frac{8}{3} = \frac{8}{3}$ s.

E1. On the circle, the sine is $-\frac{1}{2}$ at angles $-\frac{\pi}{6}$ and $-\frac{5\pi}{6}$ in $[-\pi, \pi]$. Other angles deduct by adding 2 $k\pi$ and are out of the range requested.

E2. The cosine is at least $\frac{\sqrt{2}}{2}$ on the arch around angle 0, between $-\frac{\pi}{4}$ and $\frac{\pi}{4}$, including bounds. All the solutions are therefore $[-\frac{\pi}{4}, \frac{\pi}{4}]$.

E3. By composition, $(\cos(3x - 1))' = -3\sin(3x - 1)$. For the sine square, $(\sin^2 x)' = 2\sin x \cos x$. The two derivatives are defined on $\mathbb{R}$.

E4. $f'(x) = \cos x$, positive on $[0, \frac{\pi}{2}[$ and $]\frac{3\pi}{2}, 2\pi]$negative on $]\frac{\pi}{2}, \frac{3\pi}{2}[$. Thus f growth from 2 to 3, down from 3 to 1, then increases from 1 to 2. The maximum is 3 in $\frac{\pi}{2}$ and minimum 1 in $\frac{3\pi}{2}$.

E5. An early sine is $-\cos x$. So the integral is worth $[-\cos x]_0^{\pi} = 1 - (-1) = 2$. Value average over this length interval π is $\frac{2}{\pi}$.

Corrected from DM5

1. By product, $f' = \cos^2 x - \sin^2 x = \cos 2x$. In the range studied, 2 $x \in [0, \pi]$: the derivative is positive before $x = \frac{\pi}{4}$ and negative afterwards. The maximum is $f(\frac{\pi}{4}) = (\frac{\bar{2}}{2})^2 = \frac{1}{2}$.

2. Thanks to $\sin 2x = 2\sin x \cos x$, $f(x) \geq \frac{1}{4}$ is equivalent to $\sin 2x \geq \frac{1}{2}$. For 2 $x \in [0, \pi]$, it gives $2x \in [\frac{\pi}{6}, \frac{5\pi}{6}]$, therefore $x \in [\frac{\pi}{12}, \frac{5\pi}{12}]$.

3. The derivative of $\frac{1}{2}\sin^2 x$ is $\sin x \cos x = f(x)$. So $\int_0^{\frac{\pi}{2}} f(x)\,dx = [\frac{1}{2}\sin^2 x]_0^{\frac{\pi}{2}} = \frac{1}{2}$. Value average est $\frac{\frac{1}{2}}{\frac{\pi}{2}} = \frac{1}{\pi}$.

4. $A(a) = \frac{1}{2}\sin^2 a$. Half of the total area is $\frac{1}{4}$, therefore $\sin^2 a = \frac{1}{2}$. On $[0, \frac{\pi}{2}]$sine is positive and Strictly increasing: this equality is equivalent to $a = \frac{\bar{2}}{2}$, then to $a = \frac{\pi}{4}$.

Corrected work in the manual or an anal

The curve of $f(x) = \cos(3x)$ suggests a symmetry with regard to the axis of the ordered and a repetition after a horizontal translation of $\frac{2\pi}{3}$. For everything $x \in \mathbb{R}$, $f(-x) = \cos(-3x) = \cos(3x) = f(x)$: f is a pair. On the other hand,

$$f\left(x + \frac{2\pi}{3}\right) = \cos(3x + 2\pi) = \cos(3x) = f(x).$$

The number $\frac{2\pi}{3} > 0$ is therefore a period of f. The drawing suggests these properties; the identities establish for all real.